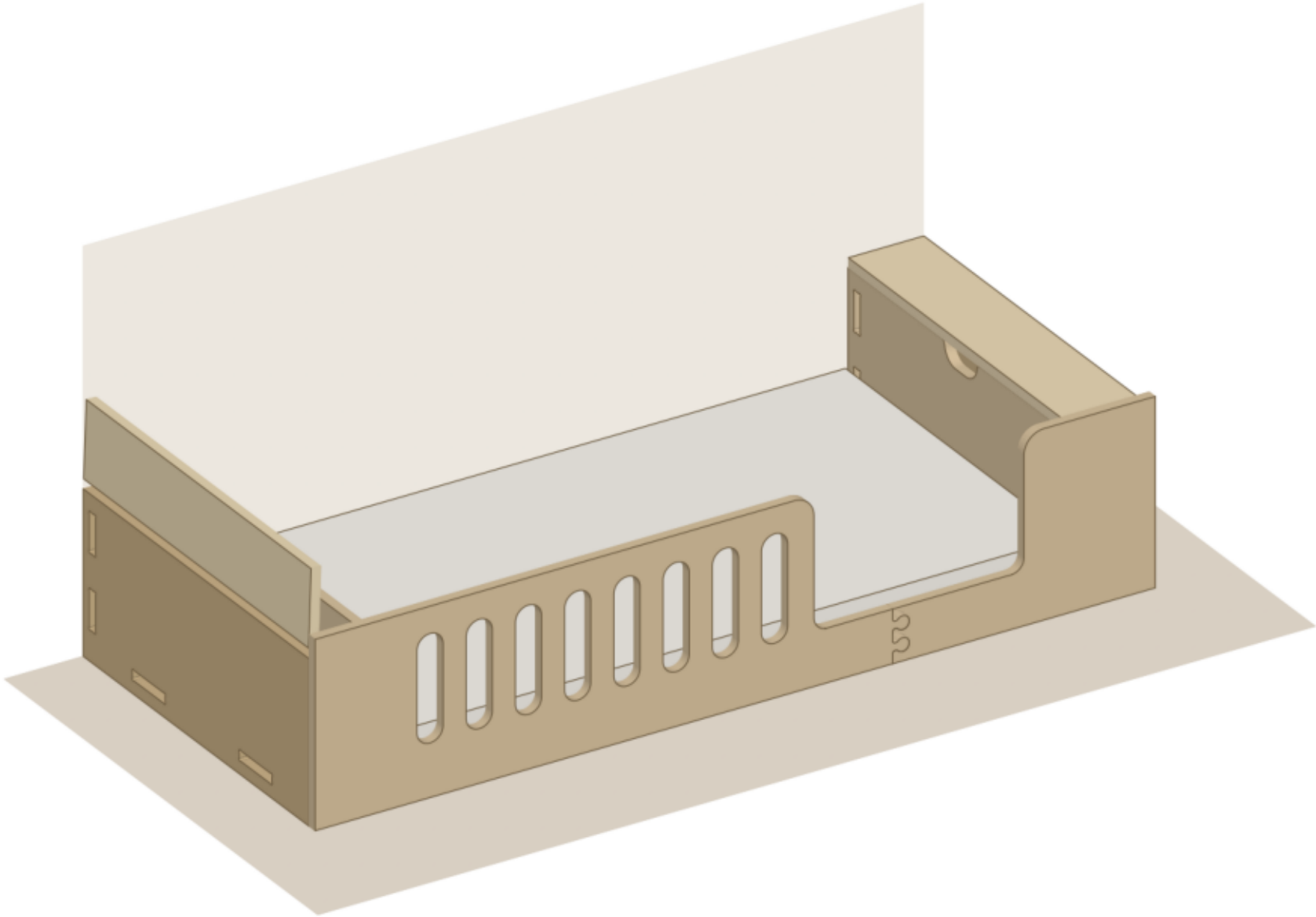
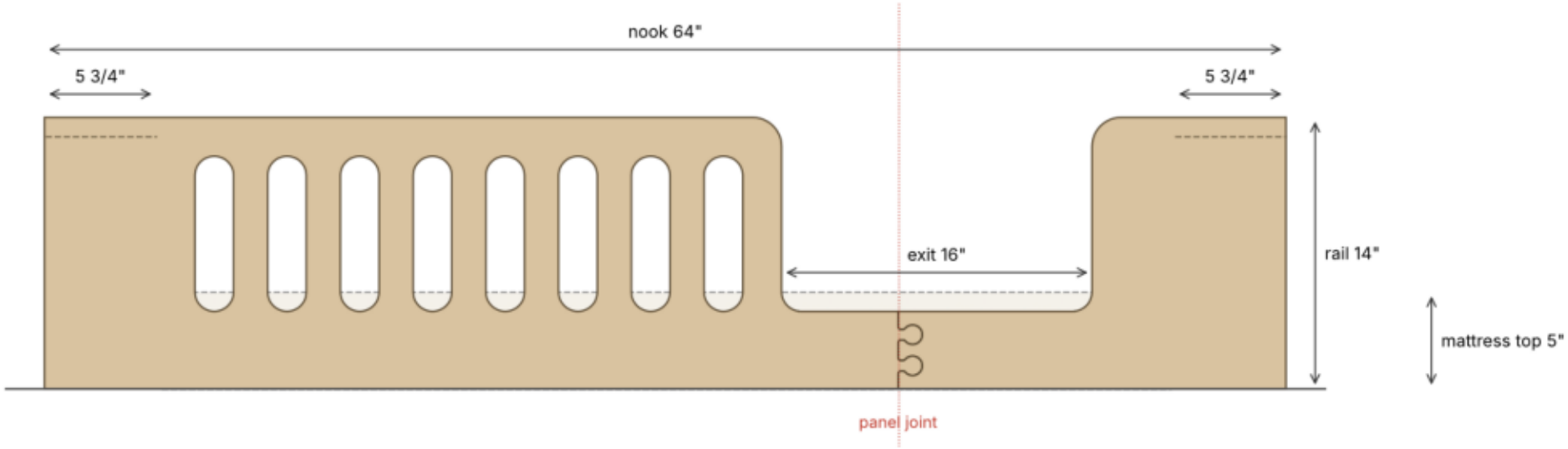
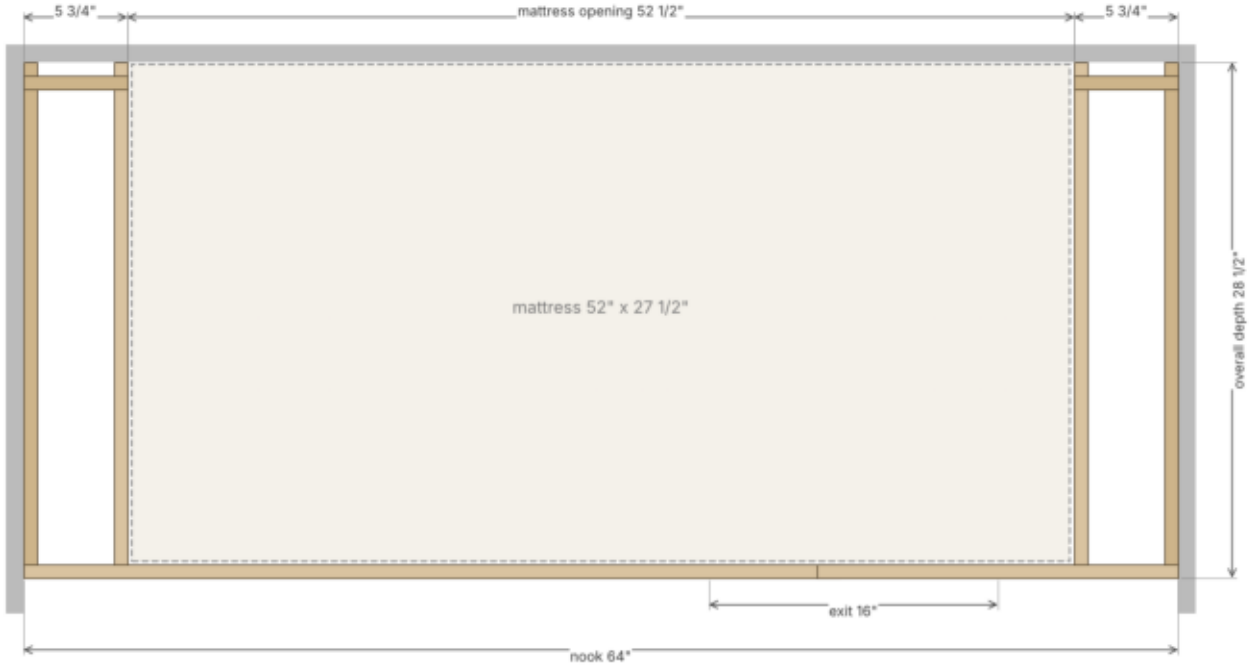
Floor bed, side exit



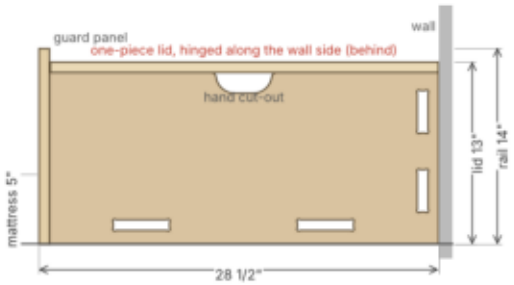
Floor bed, side exit: front view



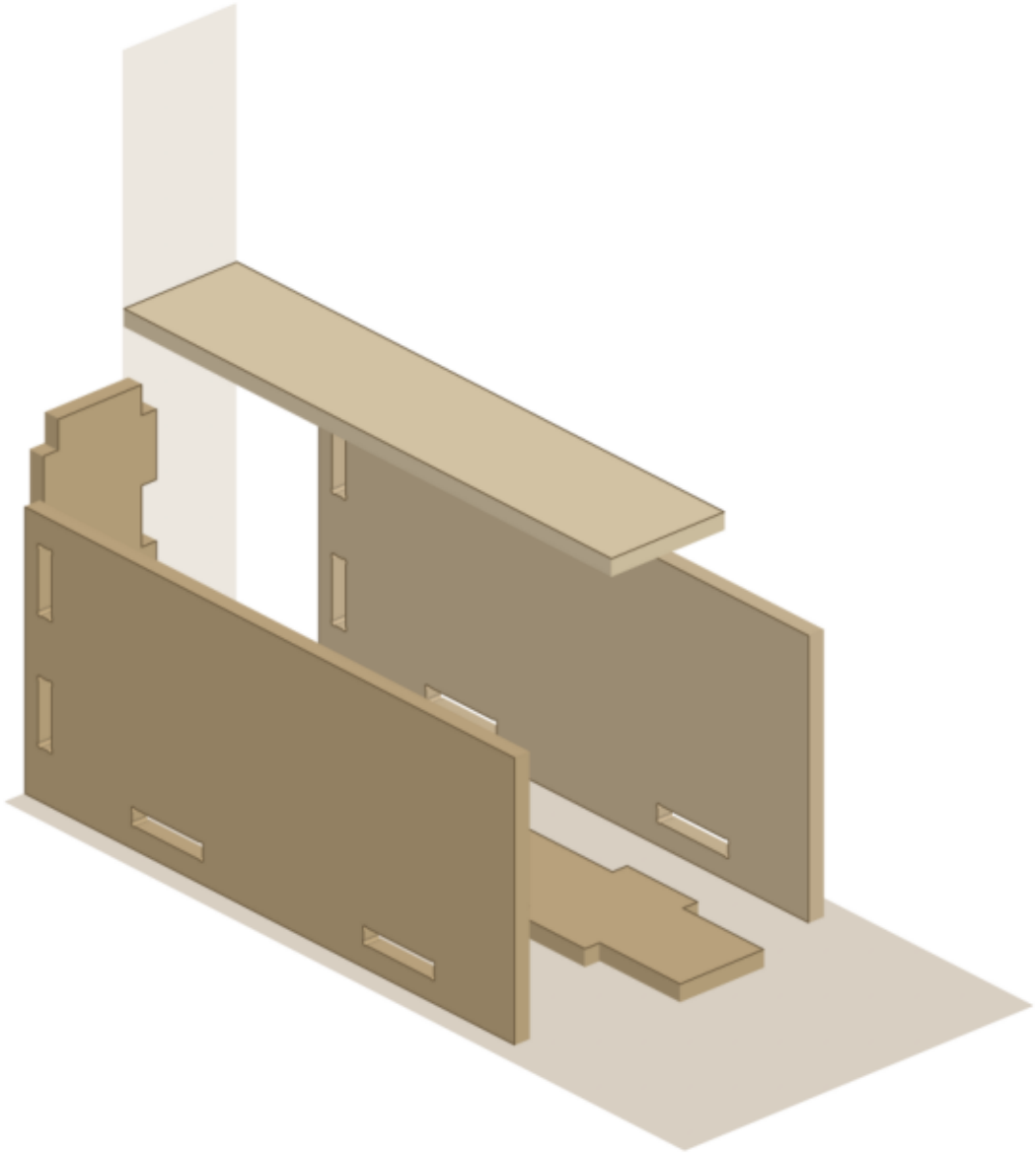
Plan (front of bed at the bottom)



Head box seen from the mattress

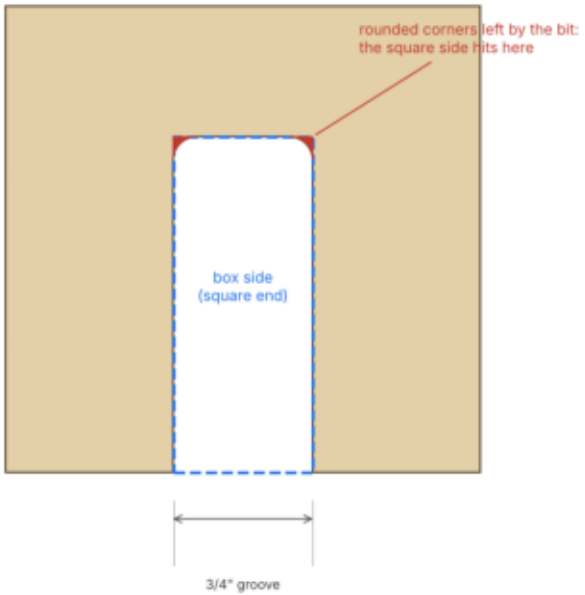


Head box, exploded (tabs pass through the sides)

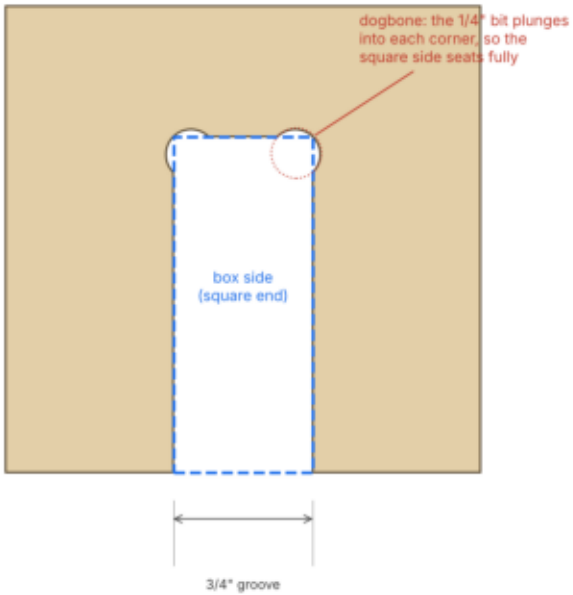


Dogbones: how a square part fits a CNC-cut groove (looking at the back of the front panel)

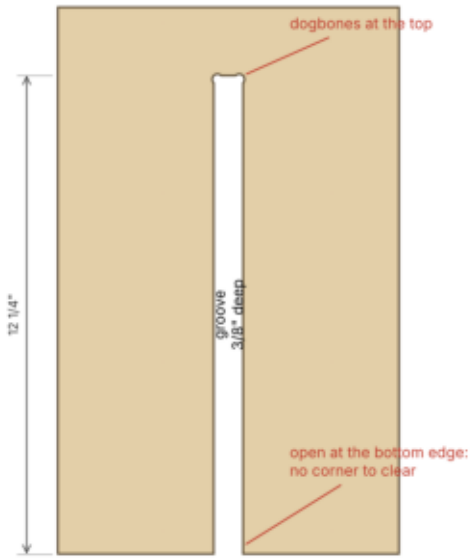
Without dogbones (red = wood that blocks the square corner)



With dogbones

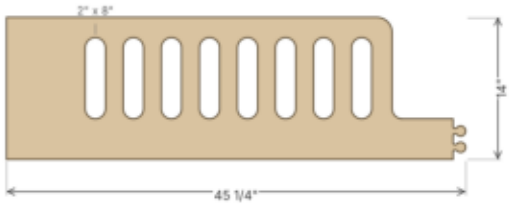


Back of the front panel, one groove

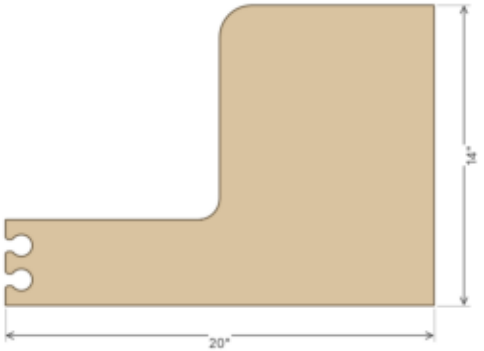


Floor bed, side exit: part drawings (inches, 3/4" plywood)

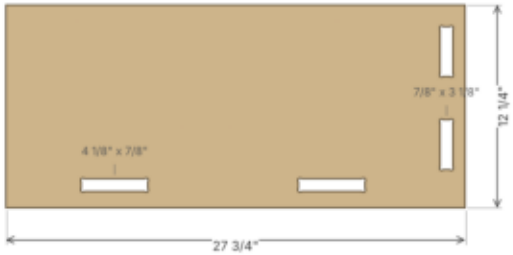
Front panel A (qty 1)
knob half



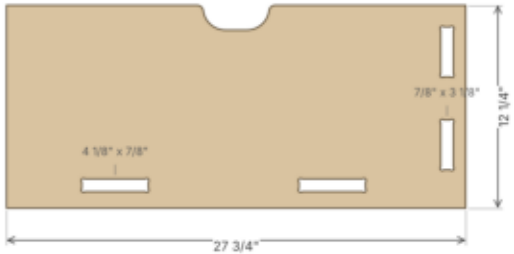
Front panel B (qty 1)
socket half



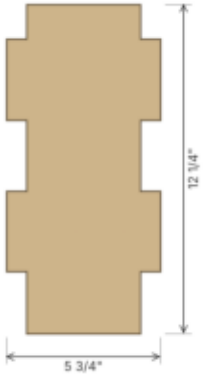
Box side (wall) (qty 2)
against the nook wall, 1 per box



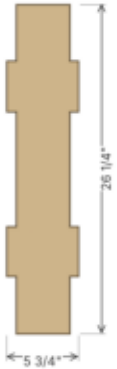
Box side (mattress) (qty 2)
faces the mattress; hand cut-out to open the lid



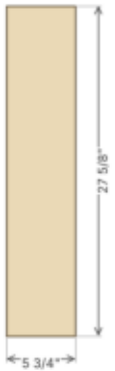
Box back (qty 2)
screw through into wall studs



Box floor (qty 2)
sits 1" off the floor

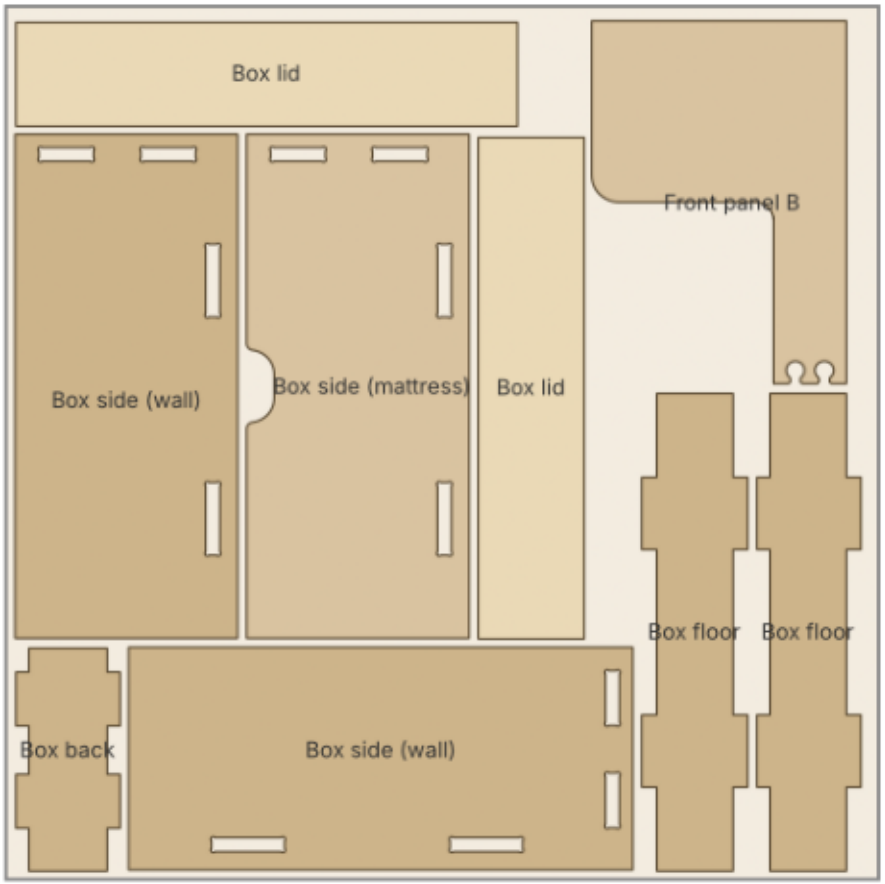


Box lid (qty 2)
piano hinge along the long wall-side edge

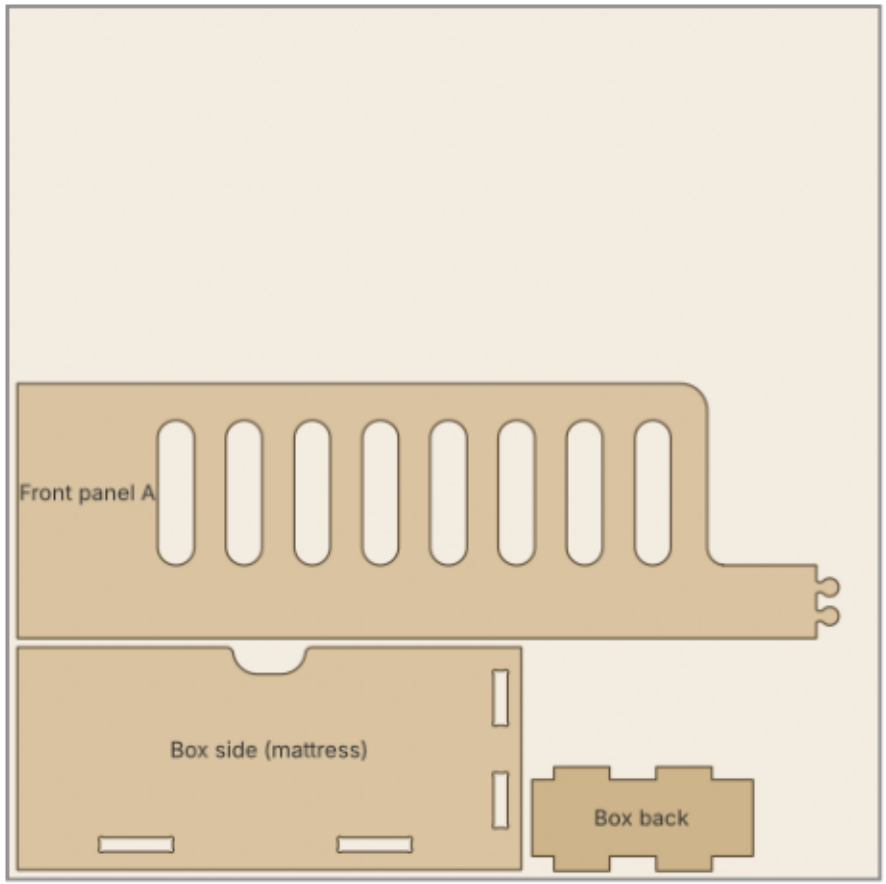


Floor bed, side exit: nesting on 48" square panels

4'x4' panel 1



4'x4' panel 2



Floor bed, side exit: cut list, hardware, assembly

Cut list

Part	Qty	Size (W x H)	Notes
Front panel A	1	45 1/4" x 14"	knob half
Front panel B	1	20" x 14"	socket half
Box side (wall)	2	27 3/4" x 12 1/4"	against the nook wall, 1 per box
Box side (mattress)	2	27 3/4" x 12 1/4"	faces the mattress; hand cut-out to open the lid
Box back	2	5 3/4" x 12 1/4"	screw through into wall studs
Box floor	2	5 3/4" x 26 1/4"	sits 1" off the floor
Box lid	2	5 3/4" x 27 5/8"	piano hinge along the long wall-side edge

Key dimensions

Panels: 2 pieces of 4'x4' (1 x 4'x8' sheet(s) of 3/4" plywood, each cut in half)
Heights from floor: mattress top 5", rail top 14", box lids 13", exit sill 4"
Exit opening: 38" to 54" from the head wall; panel halves join at 44"
Head/foot boxes: 5 3/4" wide outside, 4 1/4" inside, 27 3/4" deep

Hardware

- Wood glue (PVA)
- 1-1/4" wood screws x 16 (box joinery)
- 1-1/4" pocket-hole screws x 12 (guard panel to boxes)
- 2-1/2" screws x 4 into studs (box backs)
- Piano hinge 1-1/2" open width, 2 x 24" lengths (1 per lid)
- Soft-close lid stays x 2 (light-duty toy-box type)
- Shims as needed

Assembly

1. Cut all parts (sheets/panel_*.dxf). Profile CUT_INSIDE before CUT_OUTSIDE, use tabs/onion skin on the small parts, and dry-fit the puzzle joint on scrap first; adjust knob_clearance if needed.
2. Ease every exposed edge with a 1/4" roundover (rail top, slots, exit opening, lid edges and the hand hole). Leave the joinery edges and the faces that meet the wall square.
3. Boxes: glue the back and floor tabs into one side, add the second side, clamp square. Tabs are flush with the outside faces. Pre-drill and screw (1-1/4") through the side into each tab.
4. Set both boxes in the nook, tight to the side walls. Shim level, then screw through each box back into a stud (2-1/2" screws, 2 per box). The 3/4" behind the back lets you scribe the sides to the wall.
5. Glue the two guard-panel halves together on a flat surface (puzzle joint), let it cure.
6. Fix the guard panel to the front edges of the box sides with glue + 1-1/4" pocket screws (3 per